

Kernel Obstructions for Quantum Integration and a Regular Scalar-Injective POVM

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Status: candidate partial result, likely valid. Two broad density regimes force noninjectivity of the full quantum integration map. Conversely, an explicit nonatomic POVM with norm-continuous uniformly coercive density has injective scalar integration, although its full quantum integration map is noninjective and its range is nonconvex. The general injectivity question remains open only in a sharply isolated singular-density regime.

1 Source problem and setup

Plosker and Ramsey [1] define, for a POVM $\nu : \Sigma \rightarrow \mathcal{B}(\mathcal{H})$ and a full-rank density operator ρ , the scalar measure

$$\mu = \nu_\rho, \quad \mu(E) = \text{Tr}(\rho\nu(E)),$$

and the operator-valued Radon–Nikodym density $D = d\nu/d\mu$, when it exists. Their quantum integration map is

$$\mathcal{E}_\nu : L^\infty_{\mathcal{H}}(X, \nu) \longrightarrow \mathcal{B}(\mathcal{H}), \quad \mathcal{E}_\nu(F) = \int_X F \, d\nu,$$

characterized ultraweakly by

$$\text{Tr}(s\mathcal{E}_\nu(F)) = \int_X \text{Tr}\left(sD(x)^{1/2}F(x)D(x)^{1/2}\right) d\mu(x) \quad (s \in \mathcal{T}(\mathcal{H})). \quad (1)$$

On arXiv PDF page 7 they identify “whether $\mathcal{E}_\nu$ is injective or not” as a natural problem. They then call scalar integration *classically non-injective* when, on every nonnull measurable subset, the restriction $f \mapsto \mathcal{E}_\nu(fI)$ has a nonzero kernel. This stronger local scalar condition is a hypothesis of their operator-valued Lyapunov theorem.

Throughout, $\mathcal{H}$ is separable, $\nu(X) = I$, D exists, and ν is nonatomic. Since ν and $\mu = \nu_\rho$ are mutually absolutely continuous, μ is nonatomic as well.

2 Proof intuition

Formula (1) shows that quantum integration first sandwiches $F(x)$ by $D(x)^{1/2}$ and then averages. A pointwise kernel of D therefore gives an immediate kernel element. At the opposite extreme, if D is uniformly invertible on any nonnull region, one can prescribe the sandwiched integrand there: conjugate a fixed operator A by $D^{-1/2}$ and multiply by a nonzero scalar function of mean zero.

$$\begin{aligned}
& \rightarrow \int_X \text{Tr} \left(\left(\overline{d\nu_\rho}^{(x)} \right)^\top s \left(\overline{d\nu_\rho}^{(x)} \right) \psi(x) \right) d\nu_\rho(x) \\
& = \text{Tr} \left(s \int_X \psi(x) d\nu(x) \right),
\end{aligned}$$

from which it follows that $\int_X \psi_n d\nu \xrightarrow{wk^*} \int_X \psi d\nu$. $\square$

A very natural problem that arises from the study of integration against a POVM is that of whether $\mathcal{E}_\nu$ is injective or not. More specifically, for a POVM $\nu : \mathcal{O}(X) \rightarrow \mathcal{B}(\mathcal{H})$ we say that integration against ν is *classically non-injective* if for every $E \in \mathcal{O}(X)$ such that $\nu(E) \neq 0$ we have that

$$f \in L^\infty(E, \nu) \mapsto \mathcal{E}_\nu(fI_{\mathcal{H}})$$

is non-injective.

It is immediate that integration against any measure with an atom fails this condition. By cardinality alone it is also immediate that integration against a non-atomic finite-dimensional POVM is classically non-injective.

Such a condition is quite essential to proving a convexity result as shown by an example of Uhl [3, Example 1, Chapter IX].

Example 2.9 (Uhl). *Let μ be Lebesgue measure on $[0, 1]$ and $L^\infty([0, 1], \mu) \subset \mathcal{B}(\mathcal{H})$. Define the POVM $\nu : \mathcal{O}([0, 1]) \rightarrow \mathcal{B}(\mathcal{H})$ by*

$$\nu(E) = \chi_E, \quad \forall E \in \mathcal{O}([0, 1]).$$

Then the range space $\mathcal{R}_\nu = \{\nu(E) : E \in \mathcal{O}([0, 1])\}$ is not convex. Note that it can be proven that this non-atomic measure is not classically non-injective.

Figure 1: The injectivity problem and the classically non-injective condition on source PDF page 7.

This leaves only one possible regime for injectivity: $D(x)$ must be injective but arbitrarily small in some directions almost everywhere. The explicit circle example below shows why the scalar version does not settle the full question. Its off-diagonal blocks record every Fourier coefficient, so scalar integration is injective; nevertheless uniform invertibility lets the operator-valued cancellation above work.

3 Two kernel criteria

Theorem 3.1 (Pointwise-kernel criterion). *Suppose*

$$K = \{x \in X : \ker D(x) \neq \{0\}\}$$

has positive μ -measure. Then $\mathcal{E}_\nu$ is not injective.

Proof. Let $P(x) = \mathbf{1}_{\{0\}}(D(x))$ on K , and put $P(x) = 0$ off K . This is a weakly measurable projection field: for example it is the strong pointwise limit of $\mathbf{1}_K(x)(I + nD(x))^{-1}$. It is essentially bounded, and it is nonzero in $L^\infty_{\mathcal{H}}(X, \nu)$ because $\mu(K) > 0$ and $\nu \sim \mu$. Pointwise,

$$D(x)^{1/2}P(x)D(x)^{1/2} = 0.$$

Equation (1) therefore gives $\mathcal{E}_\nu(P) = 0$. $\square$

Theorem 3.2 (Local-coercivity criterion). *Suppose there are $E \in \Sigma$ and $c > 0$ such that $\mu(E) > 0$ and*

$$D(x) \geq cI \quad \text{for } \mu\text{-almost every } x \in E.$$

Then $\mathcal{E}_\nu$ is not injective. More precisely, for every nonzero $A \in \mathcal{B}(\mathcal{H})$ there is a nonzero $F \in L^\infty_{\mathcal{H}}(X, \nu)$ with $\mathcal{E}_\nu(F) = 0$.

Proof. By nonatomicity, split E into measurable sets E_1, E_2 of positive μ -measure and set

$$g = \mu(E_2)\mathbf{1}_{E_1} - \mu(E_1)\mathbf{1}_{E_2}.$$

Then g is nonzero, supported in E , and $\int_E g d\mu = 0$. Functional calculus gives $\|D(x)^{-1/2}\| \leq c^{-1/2}$ on E . Define

$$F(x) = \begin{cases} g(x)D(x)^{-1/2}AD(x)^{-1/2}, & x \in E, \\ 0, & x \notin E. \end{cases}$$

This is measurable and essentially bounded, is nonzero, and satisfies

$$D(x)^{1/2}F(x)D(x)^{1/2} = g(x)A.$$

Thus $\mathcal{E}_\nu(F) = A \int_E g d\mu = 0$. □

Proposition 3.3 (Necessary shape of an injective density). *If $\mathcal{E}_\nu$ is injective, then for μ -almost every x , the positive operator $D(x)$ is injective and has dense, nonclosed range.*

Proof. Theorem 3.1 gives $\ker D(x) = 0$ almost everywhere, hence $\overline{\text{ran } D(x)} = \mathcal{H}$. If $D(x)$ were bounded below on a positive-measure set, then one of the measurable sets $\{x : D(x) \geq n^{-1}I\}$ would have positive measure, contradicting Theorem 3.2. Thus $D(x)$ is not bounded below almost everywhere. An injective positive operator with closed range is bounded below, so its range here is nonclosed. □

4 A regular scalar-injective example

Theorem 4.1 (Fourier POVM). *There is a nonatomic quantum probability measure ν on the circle with a norm-continuous Radon–Nikodym density satisfying*

$$\frac{3}{4}I \leq D(z) \leq \frac{5}{4}I,$$

such that:

1. *scalar integration $T(f) = \mathcal{E}_\nu(fI)$ is injective on $L^\infty(\mathbb{T})$, so ν is not classically non-injective;*
2. *the full quantum integration map $\mathcal{E}_\nu$ is noninjective;*
3. *the range $\{\nu(E) : E \in \Sigma\}$ is not convex.*

Proof. Let $\mathcal{H} = \bigoplus_{k \geq 1} \mathbb{C}^2$, with basis $e_{k,0}, e_{k,1}$ in the k th block. Put $a_k = 2^{-k-1}$ and, for $z \in \mathbb{T}$, define the block diagonal operator

$$D(z)|_{\mathbb{C}_k^2} = \begin{pmatrix} 1 & a_k z^k \\ a_k \bar{z}^k & 1 \end{pmatrix}.$$

Each block has eigenvalues $1 \pm a_k$, so $\frac{3}{4}I \leq D(z) \leq \frac{5}{4}I$. Since $a_k \rightarrow 0$, the block tail is uniformly small and the finitely many leading blocks are continuous; hence $z \mapsto D(z)$ is norm-continuous. With m normalized Haar measure,

$$\int_{\mathbb{T}} D(z) dm(z) = I.$$

Define $\nu(E) = \int_E D(z) dm(z)$.

Take the full-rank density operator

$$\rho|_{\mathbb{C}_k^2} = 2^{-k-1}I_2.$$

Its trace is $\sum_{k \geq 1} 2 \cdot 2^{-k-1} = 1$, and the diagonal form of ρ gives $\text{Tr}(\rho D(z)) = 1$. Therefore $\nu_\rho = m$ and $D = d\nu/d\nu_\rho$. Mutual absolute continuity with nonatomic m shows that ν is nonatomic.

For $f \in L^\infty(\mathbb{T})$, the diagonal entries of $T(f)$ equal $\int f dm$, while the two off-diagonal entries in block k equal

$$a_k \int_{\mathbb{T}} f(z) z^k dm(z), \quad a_k \int_{\mathbb{T}} f(z) \bar{z}^k dm(z).$$

Thus $T(f) = 0$ forces every Fourier coefficient of f to vanish. Since $f \in L^1(\mathbb{T})$, Fourier uniqueness gives $f = 0$ almost everywhere. Hence T is injective, and so is its restriction to functions supported on any measurable subset.

On the other hand, Theorem 3.2 applies globally, proving that $\mathcal{E}_\nu$ is noninjective. Explicitly, for any nonzero $A \in \mathcal{B}(\mathcal{H})$,

$$F(z) = zD(z)^{-1/2}AD(z)^{-1/2}$$

is bounded and nonzero, while $\mathcal{E}_\nu(F) = A \int_{\mathbb{T}} z dm(z) = 0$.

Finally, if the range of ν were convex, it would contain the midpoint $I/2$ of $0 = \nu(\emptyset)$ and $I = \nu(\mathbb{T})$. Thus some measurable E would satisfy

$$T(\mathbf{1}_E) = \nu(E) = \frac{1}{2}I = T(\frac{1}{2}).$$

Injectivity of T would force $\mathbf{1}_E = 1/2$ almost everywhere, which is impossible. Hence the range is not convex. $\square$

5 Limitations, upgrade attempts, and novelty

The results do not decide whether an injective $\mathcal{E}_\nu$ exists in the only remaining regime: $D(x)$ injective with dense nonclosed range almost everywhere. Six focused upgrade attempts included inverse-density kernels, measurable kernel projections, a commutator cancellation formula, a Stinespring compression reformulation, a predual density reformulation, and a search for an injective Fourier model. The commutator and inversion routes become unbounded exactly in the remaining regime; the functional-analytic reformulations supplied no contradiction. The Fourier search instead produced Theorem 4.1, which sharply separates scalar from full injectivity.

A bounded search through 11 August 2026 checked the run indexes, local arXiv corpus, exact source title and id, and web/arXiv queries combining “classically non-injective”, “POVM integration injective”, “operator-valued Lyapunov”, and “Radon–Nikodym density”. It found the source and general informationally complete POVM literature, but no later resolution of the stated injectivity problem and no occurrence of the block Fourier construction above. The novelty verdict is therefore provisional pending expert review.

6 Human-review recommendation

The partial result is recommended for expert review. The main checks are: measurability of the kernel projection in Theorem 3.1, domain membership of the inverse-density kernel in Theorem 3.2, and the Fourier uniqueness identification in Theorem 4.1. The packet should not be promoted to a full solution unless the singular injective-density regime is settled.

References

- [1] S. Plosker and C. Ramsey, *An operator-valued Lyapunov theorem*, J. Math. Anal. Appl. **469** (2019), 117–125; arXiv:1801.03109.